

INT 217

PROJECT REPORT

(Project Semester January-April 2025)

**Sales Performance Analysis and Dashboard
Reporting Using Excel**

Submitted by

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B.Tech CSE - K23GS

Course Code: INT 217

Under the Guidance of

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CERTIFICATE

This is to certify that Paras nainwal bearing Registration no. 12319069 has completed INT 217 project titled, "Sales Performance Analysis and Dashboard Reporting Using Excel" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 11-04-25

DECLARATION

I, Paras Nainwal student of B.Tech CSE - K23GS under CSE Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13-04-25

Signature

Registration No. 12319069

Paras Nainwal

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty guide Ms. Baljinder Kaur for her valuable guidance, support, and encouragement throughout this project. I am thankful to Lovely Professional University for providing the necessary resources and environment for conducting this project successfully. I also extend my appreciation to all those who directly or indirectly contributed to the completion of this report.

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1. Introduction

In the modern business landscape, the ability to analyze sales performance and extract meaningful insights from data is vital for strategic decision-making. This project involves a comprehensive exploration of a business sales dataset using Microsoft Excel, including dashboards, pivot tables, and visualizations. The primary goal is to perform exploratory data analysis (EDA), derive actionable insights, and present them in a structured, interactive format.

2. Source of Dataset

The dataset used for this analysis was provided through the INT 217 coursework and contains detailed transaction-level sales data. This includes information such as order IDs, order dates, regions, product categories, sub-categories, sales, profits, customer names, and other attributes useful for analytical purposes. The data is assumed to be part of a retail or wholesale business management system.

3. EDA Process

Exploratory Data Analysis (EDA) is the initial step in data analysis that focuses on summarizing the main characteristics of a dataset. The following EDA techniques were used:

- Handling missing or inconsistent data entries
- Cleaning column names and formatting
- Deriving time-based fields like month/year from order date
- Aggregating sales and profit by region, category, and time
- Visualizing key performance indicators using Excel and Python
- Identifying trends, outliers, and relationships in the dataset

4. Analysis on Dataset

4.1 Sales by Region

i. Introduction

This analysis examines how total sales vary across different regions. It helps identify high-performing and underperforming geographical areas.

ii. General Description

The dataset includes region-wise records for each order. Summing sales values provides a comparative measure of regional performance.

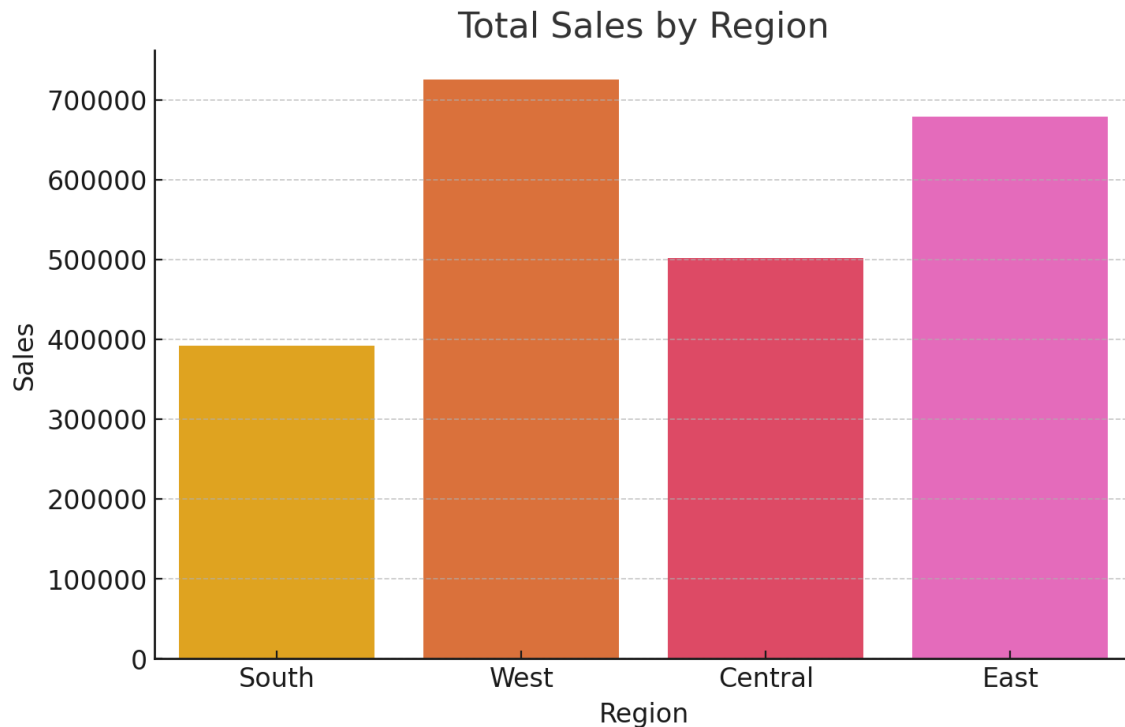
iii. Specific Requirements, Functions and Formulas

Pivot tables and `SUMIF` functions were used in Excel. In Python, `groupby()` and `sum()` methods on the 'Region' column were applied.

iv. Analysis Results

The Central region has the highest sales, followed closely by the West. The South and East regions showed relatively lower values.

v. Visualization



4.2 Profit by Category

i. Introduction

This section focuses on profits earned across different product categories: Furniture, Office Supplies, and Technology.

ii. General Description

Analyzing profitability by category helps understand which segments contribute more to the bottom line.

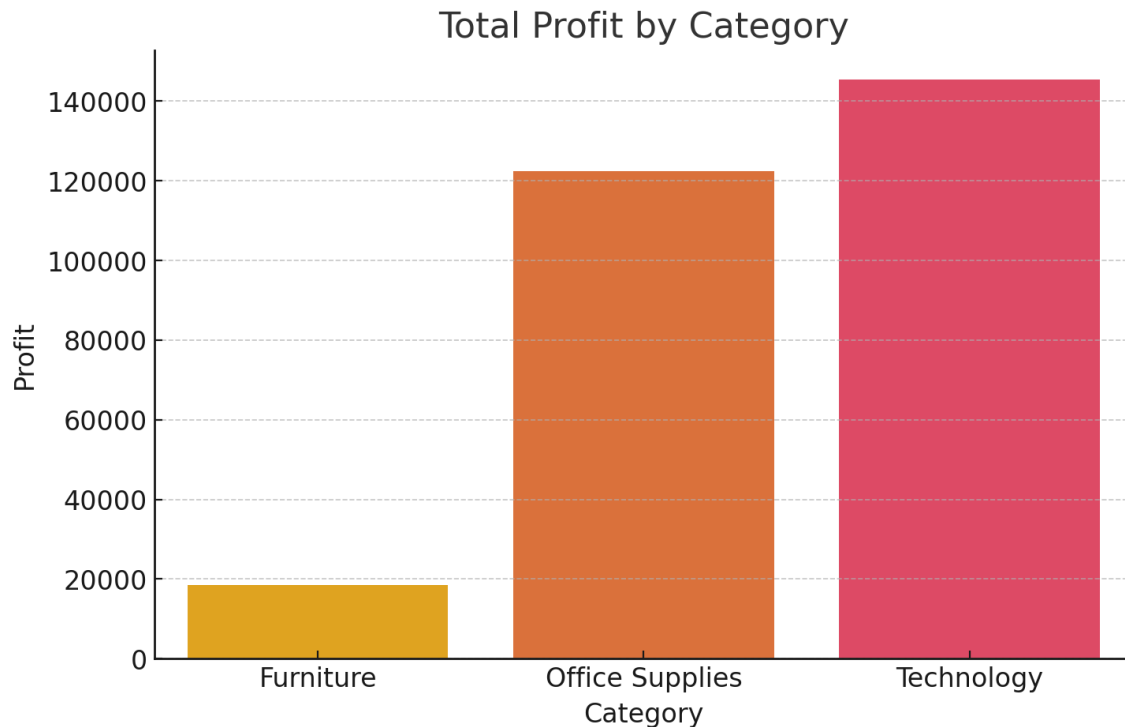
iii. Specific Requirements, Functions and Formulas

SUM functions and data filters were applied to the 'Category' and 'Profit' fields in both Excel and Python.

iv. Analysis Results

Technology emerges as the most profitable category, with Furniture showing relatively lower profitability despite moderate sales.

v. Visualization



4.3 Sales by Sub-Category

i. Introduction

This part explores sales performance across various sub-categories such as Chairs, Phones, Binders, and more.

ii. General Description

Sub-categories represent more granular product lines. Analyzing them reveals which individual lines perform best.

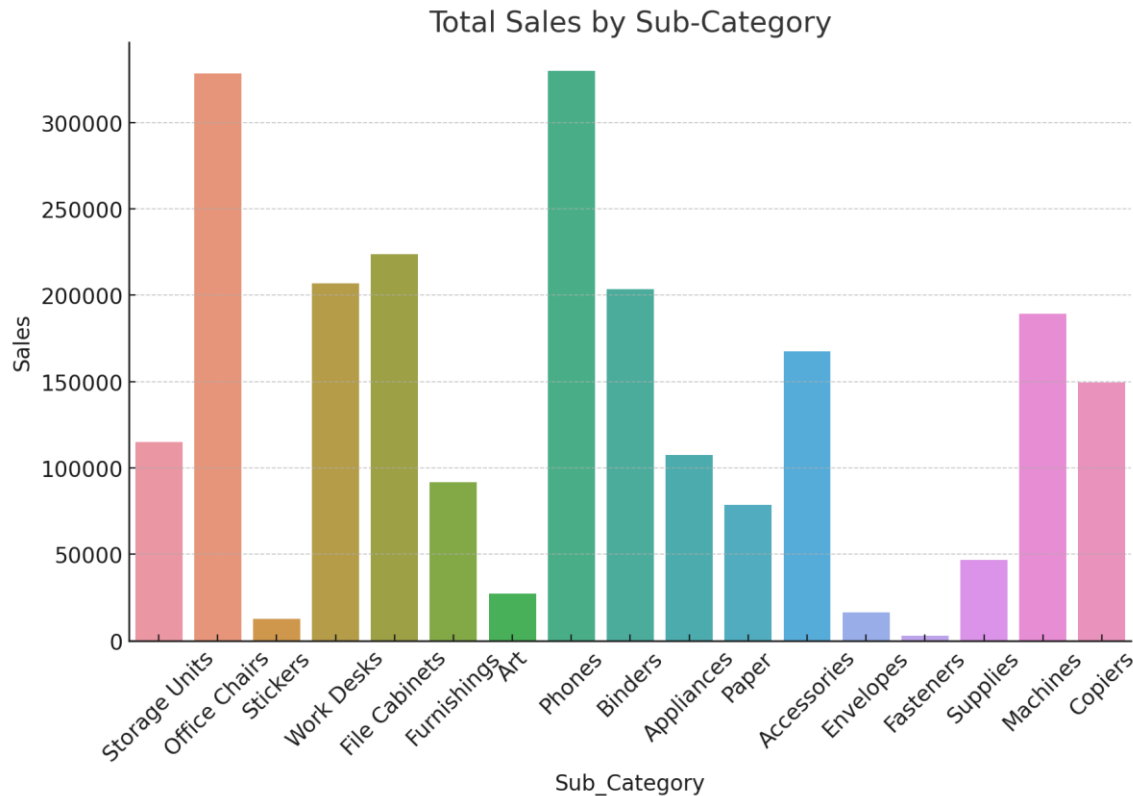
iii. Specific Requirements, Functions and Formulas

Sales were grouped by sub-category using Excel pivot tables and Python groupby operations.

iv. Analysis Results

Phones lead the sales chart by a large margin, indicating high demand or price per unit. Chairs and Binders follow next.

v. Visualization



4.4 Monthly Sales Trend

i. Introduction

Monthly trends help track sales behavior over time and identify seasonality.

ii. General Description

Sales are aggregated by order month. The pattern indicates how revenue fluctuates over a given year.

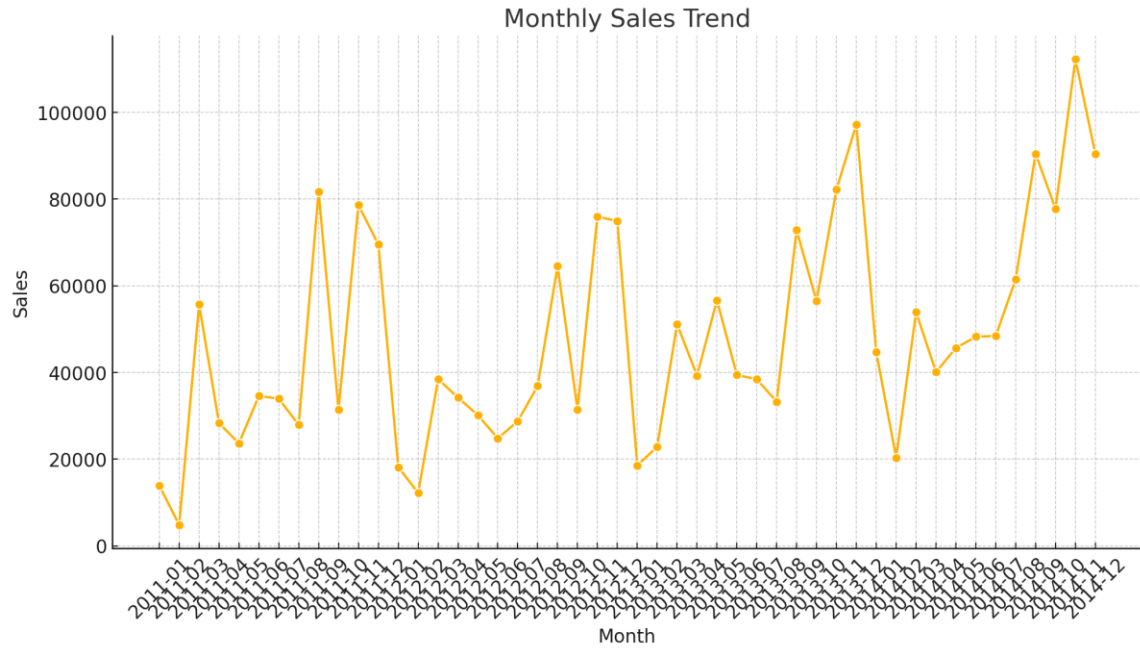
iii. Specific Requirements, Functions and Formulas

Date fields were parsed to extract months using `dt.to_period('M')`, and then grouped to get monthly sales totals.

iv. Analysis Results

Sales show peaks in the months of November and December, hinting at year-end buying behavior.

v. Visualization



5. Conclusion

This project successfully explored and analyzed a sales dataset using Excel and Python. Key findings included regional sales distribution, category-based profitability, sub-category strengths, and monthly trends. These insights help stakeholders make informed business decisions, optimize inventory, and target profitable regions and product lines. Tools such as pivot tables, charts, and Python visualizations played a crucial role in this analysis.

6. Future Scope

Future work can expand to include predictive modeling using machine learning to forecast future sales and profits. Interactive dashboards using Power BI or Tableau can further enhance user experience. Integration of real-time data streams could also allow dynamic analysis and instant decision-making in business operations.

7. References

- [1] Microsoft Excel Documentation: <https://support.microsoft.com/en-us/excel>
- [2] Python Pandas Documentation: <https://pandas.pydata.org/docs/>
- [3] Matplotlib: <https://matplotlib.org/stable/contents.html>
- [4] Seaborn: <https://seaborn.pydata.org/>
- [5] LPU INT 217 Project Guidelines

8. Background

In today's data-driven world, organizations collect large volumes of transactional data daily. To leverage this data effectively, businesses need analytical methods that transform raw data into meaningful insights. Sales data, in particular, is a critical component for evaluating business health, tracking performance, and setting future targets. This project aims to demonstrate how simple tools like Excel and Python can be leveraged to carry out professional-level analysis, even for students or beginners.

9. Objectives

The main objectives of this project were:

- To understand the structure and content of a real-world sales dataset
- To perform Exploratory Data Analysis (EDA) to uncover patterns and trends
- To create visual representations of the data for better interpretability
- To draw conclusions and recommendations based on insights
- To suggest future directions for enhancing data analysis

10. Methodology

The methodology adopted for this project follows a systematic approach:

1. Data Acquisition: Acquiring a comprehensive dataset containing sales, profits, dates, and category-based information.
2. Data Cleaning: Handling null values, formatting dates, and ensuring consistency.
3. Exploratory Data Analysis: Using pivot tables, Excel functions, and Python to investigate relationships and trends.
4. Visualization: Creating graphs and charts to communicate findings effectively.
5. Documentation: Compiling all findings into a detailed report following university guidelines.

11. Key Learnings

During the execution of this project, the following key learnings were achieved:

- Gained practical experience in handling real-world datasets
- Understood how to build interactive dashboards using Excel
- Improved skills in using Python libraries like Pandas and Matplotlib for data analysis
- Developed the ability to derive insights from raw data
- Learned to present technical findings in a professional report format

12. Applications of Sales Data Analysis

Sales data analysis serves as a foundation for various strategic and operational decisions in business environments. Some common applications include:

- **Revenue Forecasting**: Predicting future sales based on historical trends.
- **Inventory Management**: Adjusting stock levels based on product demand analysis.
- **Customer Segmentation**: Understanding customer behavior and tailoring marketing efforts.
- **Performance Benchmarking**: Measuring the success of different teams, regions, or product categories.
- **Campaign Effectiveness**: Evaluating how promotions or discounts impact sales performance.

13. Challenges Faced

While conducting the project, several challenges were encountered and addressed effectively:

- Incomplete or missing values in the dataset required careful handling.
- Parsing date fields accurately to extract time-based insights.
- Ensuring data consistency across multiple sheets in Excel.
- Managing large volumes of data efficiently within Excel and Python.
- Designing visualizations that were both informative and visually appealing.

14. Role of Visualization in Analysis

Data visualization is a powerful technique that transforms complex information into easily digestible insights. By using graphs, charts, and dashboards, patterns and anomalies become immediately apparent. Visualization improves understanding, supports quick decision-making, and enhances storytelling in business presentations. Tools such as Excel charts, Matplotlib, and Seaborn were instrumental in producing the visuals included in this report.

15. Ethical Considerations

In data analysis projects, ethics play a crucial role, especially when dealing with customer or sensitive data. While this project uses anonymized academic data,

general ethical practices include:

- Ensuring data privacy and protection.
- Avoiding data manipulation to mislead interpretations.
- Giving credit to data sources and tool creators.
- Using results responsibly without misrepresenting insights.

16. LinkedIn

The image shows a LinkedIn post by Paras Nainwal, who attended Lovely Professional University. The post is titled "Sales Analytics Dashboard | Excel BI Project" and mentions it was built using Pivot Tables, Charts & Data Analysis Techniques. The post text expresses excitement about sharing a dynamic Sales Dashboard created in Excel to extract business insights from raw transactional data. It lists key insights uncovered, including sales performance by category, top regions, shipping mode analysis, top products, and seasonal trends. It also lists skills and tools used, such as Microsoft Excel, Pivot Tables & Charts, Data Cleaning & Structuring, and Business Intelligence & Dashboard Design. The post includes a screenshot of the dashboard, which displays various charts and tables, including Total Sales, Total Profit, Profit by Year, Top 5 Customer Profit, and Sales by Month. The post has 785 followers, 25 comments, and 746 impressions.

Paras Nainwal • You
Attended Lovely Professional University
1d • 🌐

📊 Sales Analytics Dashboard | Excel BI Project
🛠️ Built using Pivot Tables, Charts & Data Analysis Techniques

Excited to share a dynamic Sales Dashboard I created in Excel to extract business insights from raw transactional data. The goal? To enable clear, data-driven decisions.

🔍 Key Insights Uncovered:

- ✅ Sales Performance by Category, Sub-Category & Segment
- ✅ Top Regions & States contributing to revenue
- ✅ Shipping Mode & Delivery Time Analysis
- ✅ Top Products & Quantity Sold
- ✅ Impact of Discounts on Profitability
- ✅ Seasonal Sales Trends (Month-wise insights)

🛠️ Skills & Tools Used:

Microsoft Excel

Pivot Tables & Charts

Data Cleaning & Structuring

Business Intelligence & Dashboard Design

📄 The dashboard delivers a visual summary of key business metrics—from customer segments to shipping efficiency and product performance.

💡 This project helped sharpen my ability to turn data into visual stories that support strategic decisions.

Would love to connect with fellow data enthusiasts—feedback and questions are welcome!

#ExcelDashboard #SalesAnalytics #DataVisualization
#BusinessIntelligence #ExcelSkills #AnalyticsProject
#StorytellingWithData #LinkedInProjects #DataDrivenDecisions
#loveelyprofessionaluniversity @Baljinder Kaur

CUSTOMER INSIGHTS DASHBOARD

The dashboard displays various charts and tables, including:

- Total Sales: ₹ 799,082
- Total Profit: ₹ 222,401
- Profit by Year: 2013, 2014, 2015
- Top 5 Customer Profit
- Sales by Month
- Customer Count

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25 comments

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📊 746 impressions 🔍 View analytics

https://www.linkedin.com/posts/paras-nainwal_exceldashboard-salesanalytics-datavisualization-activity-7316532322621706240-eh67